

## LETING, China

WMO: 545390

Lat: 39.4291N

Lon: 118.8867E

Elev: 12

StdP: 101.18

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.1      | -10.3 | -26.3                         | 0.3 | -5.5 | -24.2 | 0.4 | -5.3 | 8.1                   | -1.9 | 7.2 | -2.1 | 1.6                   | 290  | 0.418 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 6.8                    | 32.7            | 24.4 | 31.4 | 23.9 | 30.3 | 23.3 | 27.0                | 30.5 | 26.2 | 29.3 | 25.4 | 28.3 | 3.0                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 26.1                            | 21.5 | 29.2 | 25.2 | 20.4 | 28.4 | 24.5 | 19.5 | 27.6 | 85.5          | 30.7 | 81.9 | 29.5 | 78.4 | 28.3 | 29.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 7.3           | 6.2  | 5.3 | -14.6                      | 35.4 | 2.3                | 2.0 | -16.3                                              | 36.8 | -17.6      | 38.0 | -18.9      | 39.1 | -20.6      | 40.6 |    |    |
| (5)               |      |     | -15.6                      | 28.2 | 2.3                | 0.9 | -17.3                                              | 28.8 | -18.6      | 29.3 | -19.9      | 29.8 | -21.6      | 30.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 12.2        | -3.8  | -0.6  | 5.6   | 12.8  | 18.9  | 23.1  | 26.1  | 25.5  | 21.1  | 13.8  | 5.1   | -1.4  |  |  |
|                                                                        | DBStd       | 10.98       | 3.28  | 3.53  | 3.98  | 3.53  | 3.20  | 2.59  | 2.14  | 2.18  | 2.97  | 3.70  | 4.09  | 3.27  |  |  |
|                                                                        | HDD10.0     | 1404        | 429   | 298   | 147   | 13    | 0     | 0     | 0     | 0     | 0     | 10    | 152   | 354   |  |  |
|                                                                        | HDD18.3     | 2979        | 687   | 531   | 396   | 168   | 32    | 1     | 0     | 0     | 10    | 146   | 396   | 613   |  |  |
|                                                                        | CDD10.0     | 2223        | 0     | 0     | 9     | 97    | 277   | 393   | 498   | 481   | 333   | 128   | 7     | 0     |  |  |
|                                                                        | CDD18.3     | 757         | 0     | 0     | 0     | 2     | 50    | 144   | 240   | 223   | 93    | 5     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 6081        | 0     | 0     | 0     | 26    | 383   | 1049  | 2176  | 1899  | 529   | 18    | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 1837        | 0     | 0     | 0     | 3     | 85    | 323   | 741   | 585   | 99    | 1     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.4         | 2.2   | 2.5   | 2.9   | 3.1   | 2.8   | 2.5   | 2.2   | 2.0   | 2.0   | 2.2   | 2.2   | 2.1   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 638         | 3     | 6     | 10    | 27    | 42    | 79    | 224   | 164   | 45    | 22    | 11    | 4     |  |  |
|                                                                        | PrecMax     | 1163        | 17    | 33    | 33    | 120   | 170   | 269   | 482   | 355   | 132   | 60    | 48    | 26    |  |  |
|                                                                        | PrecMin     | 329         | 0     | 0     | 0     | 2     | 10    | 11    | 81    | 49    | 3     | 0     | 0     | 0     |  |  |
|                                                                        | PrecStd     | 183         | 4     | 7     | 9     | 28    | 32    | 59    | 113   | 77    | 30    | 17    | 10    | 7     |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 6.5   | 11.3  | 20.6  | 26.1  | 31.4  | 34.0  | 34.7  | 33.5  | 30.8  | 25.5  | 17.9  | 9.0   |  |  |
|                                                                        |             | MCWB        | 0.7   | 4.6   | 11.1  | 15.3  | 18.2  | 21.6  | 24.6  | 25.9  | 22.0  | 17.5  | 11.2  | 4.6   |  |  |
|                                                                        | 2%          | DB          | 4.3   | 8.7   | 17.2  | 23.4  | 28.9  | 31.6  | 33.1  | 32.0  | 29.1  | 23.3  | 15.5  | 6.8   |  |  |
|                                                                        |             | MCWB        | -0.4  | 2.9   | 9.2   | 13.8  | 17.7  | 21.5  | 25.4  | 25.1  | 20.9  | 16.2  | 10.2  | 2.4   |  |  |
|                                                                        | 5%          | DB          | 2.7   | 6.9   | 14.4  | 21.5  | 27.2  | 30.0  | 31.7  | 30.8  | 27.7  | 21.4  | 13.5  | 5.0   |  |  |
|                                                                        |             | MCWB        | -1.3  | 1.8   | 7.3   | 13.0  | 17.0  | 20.8  | 25.1  | 24.3  | 20.4  | 15.1  | 8.9   | 1.3   |  |  |
|                                                                        | 10%         | DB          | 1.3   | 5.0   | 12.1  | 19.5  | 25.5  | 28.5  | 30.4  | 29.7  | 26.4  | 19.8  | 11.6  | 3.5   |  |  |
|                                                                        |             | MCWB        | -2.0  | 0.9   | 5.8   | 11.5  | 16.0  | 20.4  | 24.5  | 24.0  | 19.8  | 14.1  | 7.7   | 0.3   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 1.7   | 5.6   | 12.6  | 17.7  | 21.1  | 24.5  | 28.3  | 28.1  | 25.1  | 19.9  | 13.2  | 5.8   |  |  |
|                                                                        |             | MCDB        | 4.6   | 9.3   | 19.0  | 23.6  | 28.3  | 29.6  | 32.0  | 31.4  | 28.2  | 23.3  | 15.5  | 7.8   |  |  |
|                                                                        | 2%          | WB          | 0.4   | 3.8   | 10.0  | 15.5  | 19.8  | 23.2  | 27.2  | 27.1  | 23.2  | 18.3  | 11.4  | 3.3   |  |  |
|                                                                        |             | MCDB        | 3.1   | 7.3   | 15.1  | 21.6  | 26.3  | 28.3  | 30.9  | 30.1  | 26.4  | 21.4  | 14.2  | 5.4   |  |  |
|                                                                        | 5%          | WB          | -0.7  | 2.6   | 8.1   | 14.0  | 18.7  | 22.4  | 26.4  | 26.2  | 22.4  | 16.9  | 9.8   | 2.0   |  |  |
|                                                                        |             | MCDB        | 2.1   | 5.9   | 13.6  | 20.0  | 24.6  | 27.3  | 29.8  | 29.1  | 25.4  | 19.6  | 12.4  | 4.1   |  |  |
|                                                                        | 10%         | WB          | -1.7  | 1.3   | 6.5   | 12.6  | 17.6  | 21.7  | 25.7  | 25.4  | 21.4  | 15.6  | 8.3   | 0.7   |  |  |
|                                                                        |             | MCDB        | 1.0   | 4.5   | 11.1  | 17.6  | 23.2  | 26.4  | 28.8  | 28.2  | 24.6  | 18.5  | 11.1  | 3.1   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.6   | 8.9   | 9.5   | 10.1  | 10.0  | 8.5   | 6.8   | 7.2   | 9.0   | 9.5   | 8.7   | 8.0   |  |  |
|                                                                        |             | MCDBR       | 10.3  | 11.3  | 12.8  | 13.2  | 12.7  | 11.3  | 8.7   | 8.4   | 10.0  | 11.0  | 11.2  | 9.7   |  |  |
|                                                                        | 5% WB       | MCWBR       | 6.7   | 6.7   | 6.6   | 6.2   | 5.1   | 4.0   | 3.4   | 3.4   | 4.0   | 5.3   | 6.7   | 6.4   |  |  |
|                                                                        |             | MCWBR       | 9.2   | 10.4  | 12.2  | 12.0  | 10.8  | 9.1   | 7.2   | 6.8   | 8.1   | 8.9   | 10.1  | 8.5   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.440 | 0.519 | 0.549 | 0.623 | 0.666 | 0.775 | 0.777 | 0.710 | 0.635 | 0.626 | 0.511 | 0.459 |  |  |
|                                                                        |             | taud        | 1.916 | 1.774 | 1.741 | 1.647 | 1.601 | 1.470 | 1.509 | 1.610 | 1.688 | 1.651 | 1.827 | 1.919 |  |  |
|                                                                        | Ebn at Noon |             | 668   | 665   | 702   | 682   | 665   | 595   | 588   | 614   | 625   | 561   | 586   | 610   |  |  |
|                                                                        |             | Edh at Noon | 138   | 183   | 210   | 245   | 262   | 299   | 285   | 251   | 219   | 205   | 151   | 129   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 2.36  | 3.11  | 4.47  | 5.27  | 5.93  | 5.32  | 4.80  | 4.68  | 4.26  | 3.22  | 2.38  | 2.05  |  |  |
|                                                                        | RadStd      |             | 0.25  | 0.31  | 0.36  | 0.27  | 0.37  | 0.49  | 0.38  | 0.31  | 0.30  | 0.27  | 0.28  | 0.19  |  |  |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | +0.69 | +0.99   | N/A    | N/A     | N/A   | N/A   | -114        | -168    | +137    | +88     |
| (47) Regional (5 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +68     | N/A     |

Nomenclature: See separate page